Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python and C++ systems**, cloud-native services, containerized workloads, and reliable data pipelines across Linux, distributed, AI, and high-throughput production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Computer Networks, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Software Engineering, Database Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked asynchronous FastAPI orchestration with explicit job states and failure isolation, processing 1K+ concurrent AI jobs at 3x throughput while sustaining **high availability, monitoring, and cloud-native reliability**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **performance analysis, data structures, and production troubleshooting**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **asynchronous execution, testing, and debugging**.
- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **REST APIs, Git, and collaborative releases**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch Decision Transformer for autonomous-driving reinforcement learning, improving planning 18% while processing 100K+ trajectories through reproducible **Linux-based evaluation pipelines, efficient algorithms, and systematic performance benchmarking**.
- Designed spatial-temporal encoders combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% through **systematic testing, numerical analysis, and documented research validation**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and large-scale data pipelines using object-oriented patterns, AWS Glue, and S3, improving response times 30% while reducing technical debt 20% across distributed production services.
- Automated 20+ unit and integration suites with PyTest, Mockito, Jenkins, and virtual machines across Windows, Linux, and macOS, strengthening **CI/CD, source control, release validation, system reliability, and maintainability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, object-oriented design, and event-driven communication** across distributed business locations.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while using Sentry, Azure Monitor, and Kibana for **observability, incident diagnosis, and reliable production releases** across multiple application environments.

Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash, HTML5, CSS3
Systems: Linux, Operating Systems, Computer Networking, TCP/IP, UDP, Concurrency, Multithreading, Algorithms, Data Structures, OOP
Cloud & DevOps: Docker, Kubernetes, AWS, Azure, GCP, Jenkins, GitHub Actions, CI/CD, Containers, Git, Terraform
Backend & Data: FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, Azure SQL, MySQL, Redis, Event-Driven Systems
Quality: Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Monitoring, Observability, Performance Profiling, Agile, SDLC, AI Tools

Projects

Subscriber Session Control Plane | *C++, Python, gRPC, Linux, Docker, Kubernetes*

- Designed a **cloud-native subscriber control plane** that authenticates sessions, applies service policies, and restores state after node failures using C++, gRPC, replicated storage, Docker, Kubernetes, and fault-injection tests.

Network Reliability Lab | *Python, Linux, TCP/IP, UDP, Prometheus, Docker*

- Built a **network performance and fault-testing toolkit** that generates TCP/UDP traffic, measures latency and packet loss, exports Prometheus metrics, and validates container recovery under congestion and service interruptions.

CPU Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, round-robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across interactive workloads through a browser-based scheduling simulator.